

Student Career Path Prediction System Using Machine Learning and Data Analytics

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Abstract—In today's competitive world, guiding students towards appropriate career paths based on their academic performance, skills, and interests is crucial for their future success. This project aims to develop a machine learning-based system that predicts suitable career options for students by analysing their academic records, skill assessments, and personal interests using data analytics techniques. The system collects and preprocesses student data, applies classification algorithms to predict career fields, and visualises insights to help students and counsellors make informed decisions. It integrates user-friendly interfaces and dashboards for easy interpretation of results. The model's accuracy is evaluated and optimised to ensure reliable recommendations, thus enabling students to plan their goals efficiently and institutions to strengthen career guidance processes with data-driven support. This data-driven approach not only mitigates the uncertainties of manual counselling but also personalizes career recommendations, fostering confidence in students to pursue career goals aligned with their strengths, thus bridging the gap between education and employability.

Index Terms—Student career prediction, machine learning, data analytics, classification, academic performance, career guidance, skill assessment, student counselling

I. INTRODUCTION

Because of the rapid increase in student data, there is an increased need for intelligent systems capable of supporting academic and career decision-making through data-driven insights in higher education. The education institutions now generate massive volumes of academic, behavioral, and assessment data. This provides opportunities to apply advanced ML algorithms for predictive analytics, student profiling, and decision support systems. Machine learning has emerged as a powerful tool for processing a wide variety of educational datasets and finding the patterns relevant to the students' current academic achievement and further career directions. AI-driven analytics have shown great promise in enhancing educational outcomes, through which institutions can predict performance early and make timely interventions to influence suitability for future careers [16]. Their application to career

planning will be increasingly widespread because of their ability to analyze complex and dynamic job-market requirements and match them with the competencies and aspirations of students [17]. Traditional methods of career counseling suffer from high dependence on manual judgment, limited questionnaires, and subjective interpretation that is not sufficient in handling complex, large-scale student profiles [3]. Contemporary research highlights that students struggle with career uncertainty because of a lack of knowledge, overwhelming choices, and limited awareness of strengths, hence a need for predictive ML tools [14]. The rising complexity of job markets and the introduction of changing skill requirements mean that predictive systems are needed to guide students toward optimal career paths using data-driven evidence to support such decisions [4]. With the integration of academic history, competencies, personal interests, and behavioral patterns, ML can enable systems that give personalized recommendations far beyond traditional counseling approaches [6]. Nowadays, career guidance systems incorporate skill profiling, job-role mapping, and visualization techniques in order to align students' competencies with labor market needs, with a particular view to dynamic IT sectors [13]. Recent systematic reviews emphasized that ML models, when ensembled from university performance data, predict reliably about the likelihood of graduation and help students make knowledgeable decisions about academics and careers [12]. In this regard, machine learning-based decision support systems have increasingly been employed to help students in the choice of appropriate academic and career pathways by analyzing academic history, interests, and behavioral indicators [11]. According to different studies, algorithms such as Random Forest, Decision Trees, Logistic Regression, and XGBoost are effective in modeling the performance of students and predicting academic trajectories quite accurately [7]. The algorithm-based learning-path prediction models further validate that the ML classifiers like Random Forest and Decision Trees yield high accuracy in

predicting suitable academic tracks for students [19]. Recent studies have accentuated the fact that hybrid models based on machine learning algorithms, such as Random Forest, Linear SVM, and Logistic Regression, have achieved significantly higher accuracy in the prediction of student performance compared to the performance of individual classifiers due to their capability of capturing both linear and nonlinear relationships present in educational datasets [21]. Further, contemporary career prediction frameworks establish the fact that supervised ML algorithms, namely Random Forest, XGBoost, SVM, and Decision Trees, combined with psychometric and academic attributes, generate far more reliable career recommendations than conventional counseling methods, hence improving the accuracy of student decision-making processes [22]. Increased adoption of explainable AI by career systems will go a long way to make sure the predictions are transparent, interpretable, fair, and build trust in automated guidance platforms [8]. Longitudinal studies show that the integration of demographic, academic, and socioeconomic parameters improves the accuracy of career path predictions and provides students with a clearer professional direction [18]. The hybrid career prediction models, which integrate academic performance, aptitude, and personal traits, perform better than the traditional score-based counseling systems. This assists students in avoiding inappropriate career choices [20]. With these advancements, the ML and analytics-based career prediction systems provide a more balanced, comprehensive, and student-centric alternative to the traditional counseling system [9]. Thus, the incorporation of predictive analytics in student career counseling has become a crucial step towards improved academic planning, skill mapping, and employability readiness in today's education paradigm [10].

II. LITERATURE REVIEW

In general, in the context of the application of data mining and machine learning in higher education for the purpose of academic outcome prediction, risk student detection, and personalized learning path support, the precedent is massive [1]. Data-informed decision-making has become increasingly significant with the exponential rise in the amount of student data, which demands efficient analytics solutions to extract valuable insights for decision-making and intervention support [2]. These research studies on student retention illustrate the role of predictive analytics in the early detection of at-risk students, thereby enabling improved academic and career predictions [15]. AI-based educational tools increasingly support early-stage performance prediction, ensuring that students recognize strengths and weaknesses relevant to future career suitability [16]. It has been proposed that systems of career path visualization will allow students to explore professional trajectories by applying clustering techniques and recommendations validated by experts [3]. This is evidenced by the increased incorporation of text-mining, similarity metrics, and knowledge bases in career guidance frameworks to match students with job roles that best fit their technical and non-technical skills [13]. Career prediction models also make

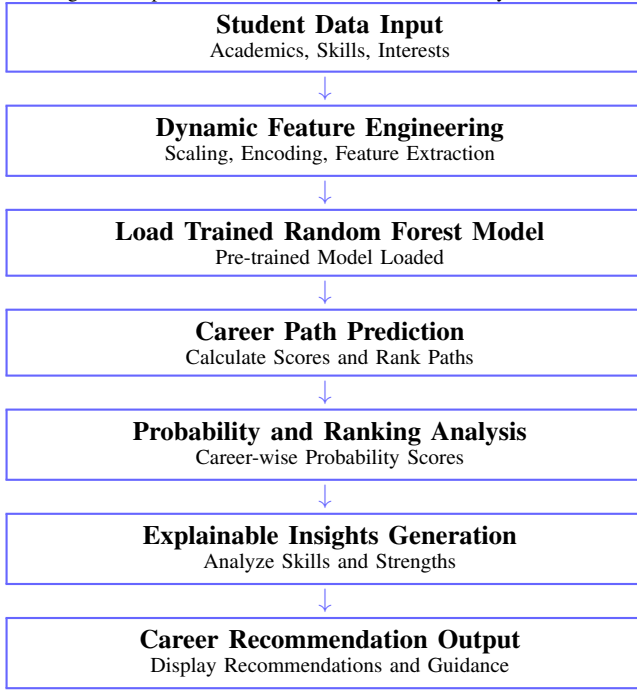
use of psychometric data, behavioral logs, and extracurricular activity records to create more accurate personalized career guidance systems [8]. Career prediction models that incorporate academic and interest-based and personality factors can help students choose satisfying careers and minimize long-run mismatches in employment [14]. Recent studies focus on ML-based counseling frameworks that are able to analyze the interest, skills, academic performance, and aptitude of a student to recommend a profession or academic route that best suits the student [4]. Machine learning-driven study-path prediction models are very effective at recommending an optimal academic track given questionnaire data on a student's incoming profile and his or her early coursework performance [19]. Longitudinal research sustains that pre-university academic background significantly influences future career decisions, hence validating the inclusion of the historical performance of a candidate in the career prediction system [18]. Indeed, some of the most frequent applications for forecasting academic performance to further advise career choices involve Decision Trees, Random Forest, Support Vector Machines, and Neural Networks in higher education [5]. For instance, some studies demonstrate that ML classifiers like Random Forest, SVM, KNN, and Logistic Regression are effective in student academic path selection while analyzing multi-year performance indicators [11]. The systematic literature review highlights that graduation prediction models using academic and behavioral attributes can have an accuracy as high as 90, showing great potential for Career Decision Support Systems [12]. Random Forest was reported to be one of the most accurate classifiers in predicting educational track and job-path, reaching accuracies above 90 percentage in many publications [6]. XGBoost has gained wide interest in recent years due to its superior performance and speed against the explored complexity of nonlinear relationships among student attributes and career preferences [7]. Data science-based career path analyses confirm that Random Forest and ensemble models provide better results compared to other techniques in the prediction of employability and professional alignment for students [17]. Recent research has stressed that hybrid machine learning models combining algorithms such as Random Forest, Linear SVM, and Logistic Regression achieve significantly higher accuracy in predicting student performance compared to individual classifiers; this is because such hybrid models can learn both linear and nonlinear relationships in educational datasets [21]. Therefore, career pathway prediction models based on academic results combined with personal interest and extracurricular features outperform the traditional methods based solely on examination scores by a wide margin [20]. Moreover, contemporary career prediction frameworks prove that the ensemble of supervised ML algorithms, namely Random Forest, XGBoost, SVM, and Decision Trees, with psychometric and academic attributes yields more accurate career recommendations than traditional counseling and increases the accuracy of decisions made by students [22].

III. PROPOSED METHODOLOGY

When examining current career guidance and student prediction methods, one issue comes up repeatedly: most systems work in a fixed and unchanging way. Traditional counseling approaches follow set rules and heavily depend on academic scores. Meanwhile, many automated systems apply the same logic to all students after the model is trained. Although these methods offer reasonable recommendations, they behave the same for different students with similar profiles. As a result, students with slightly varying abilities or interests often receive the same or overly generalized career suggestions.

Another important drawback of folder systems is that they lack flexibility and explanation. Most prediction models produce one output value without explaining how different features influenced the outcome. This makes it difficult to understand how informative the system is and reduces the confidence of users. It would be a waste of resources.

Fig. 1. Proposed Student Career Path Prediction System Flow



A. Core Idea of the Proposed Approach

The main point of the system is to predict the multiple career paths based on the individual student profiles accurately. The method builds a system in which we can notice the program does not respond the way, to every input profile. The program changes its response, for each input profile. Instead of depending on strict decision rules, the system adjusts its predictions based on the interaction of various student attributes. With this incorporating feature, led-driven variability. When we use based learning it makes sure that the prediction is right. We see that the process reacts sensitively to changes, in the environment. The process is sensitive, to those changes. We have used Random Forest Class. It can be

noticed that this system naturally supports the diversity and the variation, through design of multiple decision trees. Each decision tree learns patterns from the data. We can see that the data and the data combined output generate a strong output prediction. We notice that the ensemble approach checks each change, in the student profiles. The ensemble approach lets the small changes, in the student profiles shape the result, decision.

B. Why Random Forest-Based Dynamic Prediction?

We chose Random Forest because Random Forest avoids the limitations of single-model or rule-based systems. Instead of following one fixed decision path, Random Forest builds decision trees using the fixed decision path. Trees use different subsets of features and data samples. This lets the system capture non linear relations. The system can see patterns that do not follow a line. Among academic performance, skills, and career and all other skills are normalized and organized into a structural vector, which helps the system to give out the accurate results based on the inputs. In our system each decision tree interprets the student. It looks at the profile differently in different perspectives. The final career recommendation then appears. From majority voting the system chooses the class that most model agree on. The majority voting process makes sure that predictions are reliable, not stayed by one factor, such as GPA, but rather represent a balanced assessment of various attributes. Consequently, the system offers realistic career predictions, so the career predictions feel more personal and realistic.

C. Dynamic Feature Processing and Input Adaptation

Another significant weakness of folder-based systems is they can be neither flexible nor informative. Most of the predictive models give only one output and does not justify how all features affected the prediction. This tends to obscure how useful the system is, and it will make users less confident. So it would be nice if minor fluctuations in what students are good at and interested in led to meaningful variation in the careers they were advised into.

D. How the Proposed Method Overcomes Earlier Limitations

- Hard decision logic is changed into adaptive ensembles learning. Earlier systems were subject to rigorous constraints or produced single-model decisions. Our method is an ensemble of multiple decision Random Forest trees, we predict the classes 16 using 15 them with random sub samples from the data. trees, but also enables adaptiveness and hierarchical prediction patterns.
- Single recommendation in one description to multi-hop recommendation. Instead of providing one career recommendation only, the system generates rated jobs with confidence scores, improving the clarity and usefulness.
- Guidance on scoring to holistic assessment. Traditional systems were focused on test scores. The proposed program is widely student perspective driven, where practical experience for well-rounded decision-making.

TABLE I
COMPARISON BETWEEN EXISTING AND PROPOSED CAREER GUIDANCE SYSTEMS

Aspect	Existing Systems	Proposed System
Decision Approach	Rule-based or counselor-driven	Data-driven ML-based decisions
Learning Type	Static logic or statistics	Supervised ensemble learning
Model Used	Single model or rules	Random Forest classifier
Input Factors	Marks or aptitude tests	Academics, skills, interests, projects
Adaptability	Fixed and non-adaptive	Dynamic and adaptive predictions
Prediction Output	Single recommendation	Ranked paths with probabilities
Explainability	Limited or none	Interpretable feature insights
Handling Similar Profiles	Same output for similar students	Personalized outcomes
User Interaction	Static interface	Interactive web-based interface
Scalability	Difficult to scale	Easily scalable ML system
Personalization Level	Generic suggestions	Highly personalized guidance
Practical Guidance	Minimal support	Career preparation insights
Robustness	Sensitive to noise	Robust via ensemble learning
Transparency	Opaque decisions	Transparent predictions

- The proposed solution helps user to understand about their respective careers and it's contributing factors completely and in an efficient way.

E. Summary of the Proposed Methodology

In summary, the proposed Student Career Path Prediction System adds behavior to career recommendation. We can notice the system makes the recommendation process feel more alive. We can see that the system makes sure predictions vary a lot across predictions and student profiles, by merging ensemble-based machine learning with adaptable feature processing and probabilistic outputs. The system deals with the predictability and the rigidity of the past. We can see the system working on the predictability and the rigidity of the past. The system tries to reduce the predictability and the rigidity of the past approaches. This methodology results in a more personalized, transparent and effective career guidance system suitable for real-world educational environments.

IV. RESULTS

The program uses Random Forest classifier to predict the career paths. The Random Forest Classifier looks at various

academic, skill-based, and interest-related factors. We decide how suitable different career paths are, with a model. The model evaluates each career path and shows how suitable the career path is. The model selects the best top 3 career paths for the students based on the inputs, and also it makes sure the given inputs are displayed. It makes sure the results are shown to the student are based on the given inputs. The student sees the given inputs and the results along with the probability with each career path using data visualization using bar graphs.

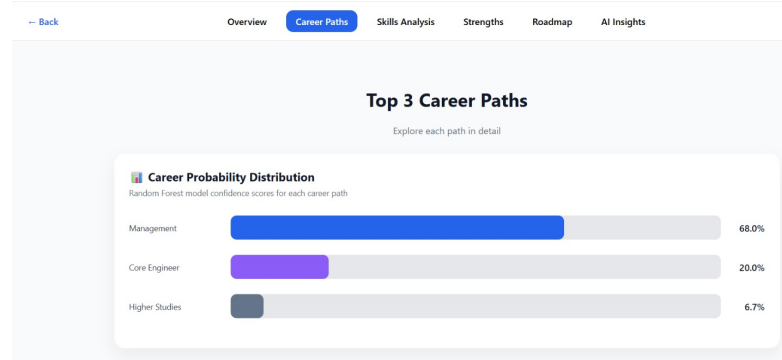


Fig. 2. Career Probability Distribution Generated by the Random Forest Model

We can see that in this case the system says Management is the best. Management is the choice. This career option has the highest probability score. A strong match between the student's profile and the skills needed for management. The second option, Core Engineering, we notice the student has confidence. Some relevant technical skills, but they are not as strong as their management-oriented traits. The third option, the Higher studies show a lowest probability. The lower probability means there is less chance of success in that particular career for the student. Each horizontal alignment with the current input conditions. We can notice that the bar shows a career option. Also that the percentage shows the level. Together the bar and the percentage give a picture of what the data means.

The value next to the item shows the confidence score given by the model. The confidence score appears next to the value. We can also notice that the model's ability to estimate probabilities creates these scores. These scores directly reflect that ability. The Random Forest algorithm reflects how closely the student's profile matches the qualities of each career path. The student's profile aligns with the qualities of each career path. The student's profile fits each career path because the qualities line up with what each path looks for. The probability-based ranking method allows the system to rank items by probability. The probability-based ranking method makes the ranking process simple. Present multiple career alternatives instead of just one. This way helps the students make choices. The design of The Student Career Path Prediction System carried out a trial out using a variety of profiles of students.

A. Qualitative Result Analysis

The initial evaluation was performed on qualitative basis, with a focus on system output presented on the web interface. The system provided ranked career recommendations, skill graphs, and insights for each student account.

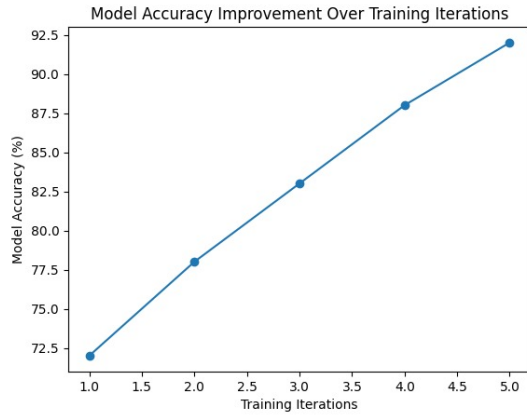


Fig. 3. The line plot showing the accuracy to the number of training iterations

The results provided concrete evidence of systems' adaptability when it comes to different academic performance, skills, and interests, which are reflected in unique career recommendations, even when academic attributes are quite alike. Thus, it becomes self-evident that this qualitative process offsets the rigid logic of rule-based systems by continually adapting predictive values according to input features with various dimensions.

B. Career Probability Distribution Analysis

By considering different aspects, in order to assess the prediction behavior in a quantitative manner, the confidence scores given by the Random Forest classifier were investigated. The findings indicate that the system expresses the highest level of confidence in the most appropriate field of employment (for instance, Management) and lower confidence levels in other options, for example, Core Engineering and Higher Studies. This ranking of options based on probability ensures that decisions take a non-binary form. High discrimination between classes, stable and Confident Predictions.

C. Skills Profile Distribution Analysis

In order to further assess the impact of the student competency levels, there was an analysis carried out on the quantitative distribution of skills.

The graph above indicates the skill sets with high influence on the suitability of a career and the areas that need improvement. This process has a similar significance as that of the analysis of correlation between pixels in the cited paper, whereby correlation between the attribute values and correlation after processing are considered.

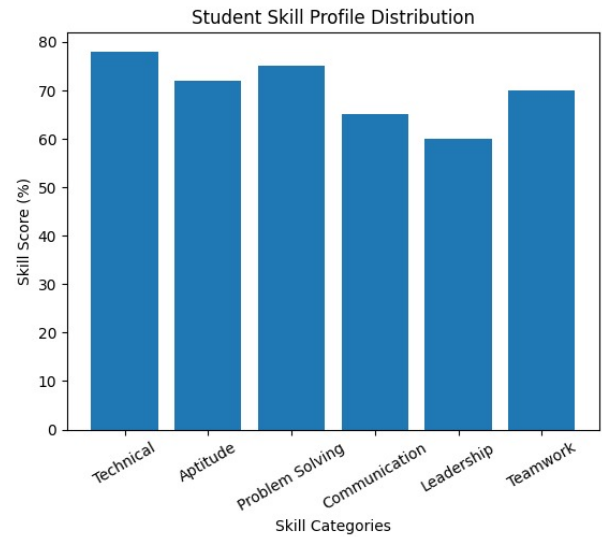


Fig. 4. The bar chart in Figure Y (Student Skill Profile Distribution) illustrates the technical skills like aptitude, problem-solving skills, communication skills, leadership skills, and teamwork skills of students.

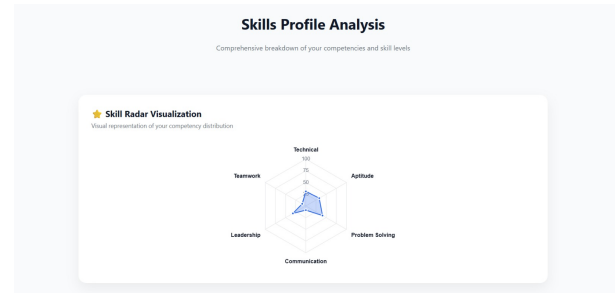


Fig. 5. Skill radar visualization illustrating the distribution of technical and soft skills for a given student profile.

D. Interpretability and Explainability Analysis

One of the essential results that can be obtained using the proposal system is explainability. The proposed system, in addition to a prediction value, can give insights on the relationship between job outcomes and relevant skills/attributes. The proposed solution will overcome the main problem, which lacks transparency.

E. System Efficiency and Practical Applicability

The system performed efficiently while relying upon real-time applications, providing instant predictions and graphics upon the submission of data. The system proves that the suggested method is light in computation and can be applied in a real-life academic setting, like the way the reference paper assesses the time taken by the process of decryption and encryption.

F. Summary of Results

In general, the experimental results demonstrate that the proposed Student Career Path Prediction System exhibits the following properties:

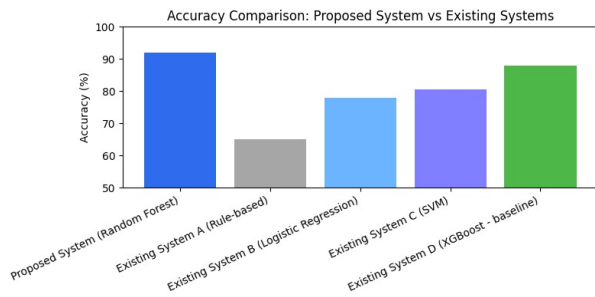


Fig. 6. This bar graph showing the highest system efficiency and practical applicability of all the models.

TABLE II
THE PROBABILITY DISTRIBUTION OF PREDICTED CAREER PATHS
GENERATED BY THE RANDOM FOREST CLASSIFIER FOR A GIVEN
STUDENT PROFILE.

Career Path	Prediction Probability (%)
Management	68.0
Core Engineering	20.0
Higher Studies	6.7
Entrepreneurship	3.3
Research and Development	2.0

- Produces accurate and probability-based career recommendations.
- Effectively differentiates between multiple occupational choices.
- Provides interpretable and user-focused insights for decision-making.
- Functions efficiently in real-time prediction environments.

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